CURRICULUM VITAE

Personal information

Name	Mewcha Amha Gebremedhin
E-mail:	mewcha96@gmail.com
ORCID	https://orcid.org/0000-0002-2356-4883
Sex	Male
Nationality	Ethiopian

About

Dr. Mewcha Gebremedhin is a hydrologist and remote sensing expert with extensive experience in applying geospatial technologies to water resource assessments. His expertise lies in integrating satellite-derived data with in-situ measurements to enhance the accuracy of rainfall, evapotranspiration, and interception loss estimates, especially in data-scarce regions. He has a strong research focus on surface-groundwater interactions and groundwater sustainability, employing integrated hydrological modeling approaches. Dr. Gebremedhin possesses advanced technical skills in GIS, remote sensing, and environmental data science, which he applies to analyze complex hydrological and ecological systems. Currently, he is actively engaged in investigating future climate change impacts on surface-groundwater interactions and salinization in coastal areas, with a specific emphasis on mitigating these effects through nature-based solutions, where such studies are critical to the resilience of communities living in coastal areas. His research extended to address the impacts of climate variability and land use changes on water resources. In his academic career, he has held leadership positions as a department head and research coordinator and contributed significantly to curriculum development. His scientific contributions have been recognized through honors like the ITC PhD publication award nomination in university of Twente, Netherlands. Dr. Gebremedhin is deeply committed to interdisciplinary collaboration to address complex challenges in global water resource management and environmental sustainability.

Education

Jul 2018-June 2024	PhD in Geosciences with a concentration in Hydrology and Water Resources from Faculty Geo-Information Science and Earth Observation, University of Twente, the Netherlands, U.S equivalent
Jan 2011-Jan 2013	Master of Science in Geoscience with a concentration in Geo-Information and Earth Observation Science for Natural Resource Management, U.S equivalent
Sep 2004-July 2006	Bachelor of Science with a double major in Computer Science and Geoscience, U.S equivalent

Professional Experience

Oct 2024- Present	Postdoctoral Research Associate at Florida Gulf Coast University (FGCU), Florida, USA
June 2019- Sep 2024	Assistant professor at Institute of Geoinformation and Earth Observation Sciences, Mekelle University, Ethiopia
Jul 2015-June 2018	Institute Council Member

Feb 2016-June 2018	Department head of Geoinformatics, Institute of Geoinformation and Earth Observation Sciences, Mekelle University, Ethiopia.
Jul 2015-Jan 2016	Research head at Institute of Geoinformation and Earth Observation Sciences, Mekelle University, Ethiopia
Jul 2015-June 2018	Lecturer at Institute of Geoinformation and Earth Observation Sciences, Mekelle University, Ethiopia.
Jan 2013 - June 2015	Lecturer, Research and Publication Expert at Adwa College of Teacher Education, Ethiopia
Dec 2008-Dec 2010	Assistance lecturer at Adwa College of Teacher Education
Sep-Nov 2008	Network Administrator at Almeda Textile Factory, Adwa, Ethiopia

Member and Teamwork

Apr 2016- June 2018	Industry-University collaboration forum member, Ethiopia
Aug –Dec 2014	Graphic Designer in Tigray Language Mother Tongue Curriculum Development Team organized by RTI International in collaboration with Ethiopian Mistry of Education.
Mar – June, 2014	BSC Developer Team for Adwa College of Teacher Education, Ethiopia

Peer-reviewed Journals and recent manuscripts

1. Birhane, E., Zenebe, M., Siyum, Z. G., Meaza, H., Abrha, H., **Gebremedhin, M. A**., Tadesse, T., Solomon, N., Beteri, J., Paso, K. G., Zhang, C., & Botelho Junior, A. B. (2026). Toxic metal contamination and remediation in artisanal and small-scale mining: An integrated global review with insights from post-conflict Tigray, Ethiopia. Manuscript draft. Submitted to Science of the Total Environment.
2. Meresa, E., Tanyanyiwa, V. I., Gumindoga, W., Chikuvired, T.J., Alavaisha, E., Abrha,H., **Gebremedhing, M.A**., Birhane, E. (2026). Spatiotemporal Land Use Land Cover Dynamics and Future Prediction using Google Earth Engine and ANN-MLP Model in the Tokwe-Mukosi Sub-Catchment, Zimbabwe. Remote Sensing Applications: Society and Environment, submitted
3. Mao, W., Ye, M., Elshall, A.S., **Gebremedhin,M.A**., Tziolasd, N. (2025). Hydrological and Biogeochemical Processes Shaping Contrasting Environmental Pathways of Chlorothalonil and Its Transformation Product SDS-3701 in Agricultural Systems. Environmental Pollution, submitted.
4. Elshall, A.S., **Gebremedhin, M.A.**, Leah L. Bremer, L.L., Burnett, K.M., Aly I El-Kad, A.I., Wada, C.A., and Ye, M. (2026). Collaborative modeling-based evaluation of groundwater sustainability in coastal aquifers using simulation optimization. Environmental Research Communications
5. **Gebremedhin, M.A**., Elshall, A.S., Ye, M, Tsegaye, S, Rotz, R. (2026). Groundwater–surface water interaction and salinity responses to projected rainfall and sea-level rise in South Florida. Environmental Research Communications
6. Elshall, A.S., Badir, A., and **Gebremedhin, M.A**. (2026). AI-affordance alignment drives authentic learning gains without foundational erosion: Quasi-experimental pilot study from an environmental data science course. Frontiers in Education.

7. **Gebremedhin, M. A.**, Daoud, M. G. and Lubczynski, M. W., (2025). Spatio-temporal variability of surface-groundwater interactions in complex hydro(geo)logical system. Journal of Hydrology, Under Review.
8. Tsegaye, S., Wise, T., Alford, G., Michael, P.R., **Gebremedhin, M.A.**, Singh, A.K , Culhane, T.H., Karatum, O., and Thomas M. Missimer, T.M .(2025). Turning Organic Waste into Energy and Food: Household-Scale Water–Energy–Food Systems. Sustainability, https://doi.org/10.3390/su17198942
9. Birhane, E., Siyum, Z. G., Shiferaw, H., Solomon, N., and Berhe, M., **Gebremedhin, M. A**. & Tesfamariam, Z. (2025). Armed conflict-driven environmental degradation and uneven socioecological resilience in Tigray, Ethiopia. Sustainable Futures, https://doi.org/10.1016/j.sftr.2025.101289
10. Solomon, N., Birhane, E., Tilahun, M. Schauer, M., **Gebremedihin, M. A**., Tikabo, F., Gidey, T. & Newete, S. W. (2024): Revitalizing Ethiopia's highland soil degradation: a comprehensive review on land degradation and effective management interventions. Discover Sustainability, https://doi.org/10.1007/s43621-024-00282-7
11. Birhane, E., Negash, E, Getachew, T., Gebrewahed, H., Gidey, E., **Gebremedihin, M. A**. (2023): Changes in total and per–capital ecosystem service value in response to land-use land–cover dynamics in north-central Ethiopia. Scientific Reports, 14, https://doi.org/10.1038/s41598-024-57151-6
12. Negash, E., Birhane, E., Gebrekirstos, A., **Gebremedhin, M. A**., Annys, S., Rannestad, M. M., ... & Nyssen, J. (2023). Remote sensing reveals how armed conflict regressed woody vegetation cover and ecosystem restoration efforts in Tigray, Ethiopia. Science of Remote Sensing, https://doi.org/10.1016/j.srs.2023.100108
13. **Gebremedhin, M. A**., Lubczynski, M. W., Maathuis, B. H., Daoud, M. G., & Teka, D. (2023). Spatio-temporal rainfall interception loss at the catchment scale from earth observation in a data-scarce area, Northern Ethiopia. Journal of Hydrology, 626, https://doi.org/10.1016/j.jhydrol.2023.130126
14. **Gebremedhin**, **M. A**., Lubczynski, M. W., Maathuis, B. H. P., & Teka, D. (2022). Deriving potential evapotranspiration from satellite-based reference evapotranspiration, Upper Tekeze Basin, Northern Ethiopia. Journal of Hydrology: Regional Studies, 41. https://doi.org/10.1016/j.ejrh.2022.101059
15. **Gebremedhin M.A**, Maciek W. Lubczynski, Ben H.P. Maathuis, Teka T. (2020). Novel approach to integrate daily satellite rainfall with in-situ rainfall, Upper Tekeze Basin, Ethiopia. Atmospheric Research, 248, https://doi.org/10.1016/j.atmosres.2020.105135
16. Woldu G., Solomon N. Hishe,H., Gebrewahid H., **Gebremedhin M.A**., Birhane E. (2020). Topographic variables to determine the diversity of woody species in the exclosure of Northern Ethiopia. Heliyon, 6(1), https://doi.org/10.1016/j.heliyon.2019.e03121
17. Birhane E., Ashfare, H., Fenta, A.A, Hishe, H., **Gebremedhin**, **M.A**, G. wahed, H., Solomon, N. (2019). Land use land cover changes along topographic gradients in Hugumburda national forest priority area, Northern Ethiopia. *Remote Sensing Applications: Society and Environment,* 13 (2019) 61–68. https://doi.org/10.1016/j.rsase.2018.10.017
18. **Gebremedhin**, **M.A**., Kahsay,G.H., Fanta, H.G. (2018). Assessment of spatial distribution of aridity indices in Raya valley, northern Ethiopia. *Applied Water Science*, 8:217 https://doi.org/10.1007/s13201-018-0868-6

19. Kahsay, G.H ,Gebreyohannes, T, **Gebremedhin, M.A**, Gebrekirstos, A, Birhane, B., Gebrewahid, H , Welegebriel, L (2018). Spatial groundwater recharge estimation in Raya basin, Northern Ethiopia: an approach using GIS based water balance model. *Sustainable Water Resources Management*, https://doi.org/10.1007/s40899-018-0272-2
20. **Gebremedhin**, M. A., Abraha, Z., & Fenta, A. A. (2017). Changes in future climate indices using Statistical Downscaling Model in the upper Baro basin of Ethiopia. Theoretical and Applied Climatology, 1–8. https://doi.org/10.1007/s00704-017-2151-4
21. Fenta, A. A., Yasuda, H., Haregeweyn, N., Belay, A. S., Hadush, Z., **Gebremedhin**, M. A., & Mekonnen, G. (2017). The dynamics of urban expansion and land use/land cover changes using remote sensing and spatial metrics: the case of Mekelle City of northern Ethiopia. *International Journal of Remote Sensing*, 38(14), 4107–4129. https://doi.org/10.1080/01431161.2017.1317936

Book and manual contribution

1. Gebremedhin, MA. (2024): Integrating in-situ data with satellite-derived products to assess surface-groundwater interactions and sustainability of groundwater resources in semi-arid environment, PhD dissertation book, https://doi.org/10.3990/1.9789036561594
2. Yonas, M., Hagos, M., Birhane, E., Tesfay, G., **Gebremedhin, MA**., Gebrehiwot, M., Gebreyesus, A., Tesfay, M., Berhe , Y., Adhana , K., Berhe, D., Hailesilassie, H. (2019). Tour Package for Southern Tigray, Ethiopia: Nature, History, Culture and Archaeology, ISBN: 9789789463969277
3. **Gebremedhin M.A (2021)**: Manual for hydrometeorological instruments installation, configuration, and data offloading

Professional award

1. Deriving potential evapotranspiration from satellite-based reference evapotranspiration, nominated to the best four papers of the year in the ITC PhD competition for publication award, 2022, University of Twente, Netherlands
2. Best action researcher, 2010, ACTE, Ethiopia

Professional Service

1. *Journal Reviewer*
    - Applied Water Science (Springer)
    - Advances in Meteorology (Wiley)
    - Atmospheric Research (Elsevier)-
    - Theoretical and Applied Climatology (Springer)
    - Water Resources Research (Wiley)
    - Environmental and Earth Sciences Proceedings, Remote Sensing (MDPI)
    - Scientific Reports (Nature Springer)
    - Remote Sensing (MDPI)
2. *Journal Guest Editor*
    - Remote Sensing (MDPI) - **Impact Factor: 4.9 ,** special issue on Remote Sensing for Ecohydrology (https://www.mdpi.com/journal/remotesensing/special_issues/SEM8JVBEB7)
3. *Conference Organization*

- AGU Meeting 2026. Generative AI in Hydrology and Interdisciplinary Environmental Research: Applications, Opportunities, and Challenges. Primary Convener. https://agu.confex.com/agu/agu26/prelim.cgi/Session/280542

Curriculum and Graduate Course Taught

- GSWM1022: Integrated Water Resources Management and Planning
- GSWM1031: Advanced Spatial Analysis and Modeling
- GSWM1023: Integrated Watershed and River Basin Management
- GSWM1032: Applications of Remote Sensing and GIS for Water Resources
- GIFM 1031: Base Data Acquisition
- GEOS1012: Principles of Database
- GEOS1011: Principles of Remote Sensing
- GEOS1012: Principles of GIS
- GIFM 1022: Introduction to GIS Programming with Python
- GIFM 1032: Image Processing
- GIFM 1031: Geo-data Visualization and Dissemination
- GSWM1021: Principles of Hydrology
- GSWM1041: Applied Geo-statistics for Water Resources
- ICT 101: Information Communication Technology

Guest Lecturer

- Urban Water Modeling Course (Graduate) at Florida Gulf Coast University: Hydrology with focused Risks, Models, and Resilience - 23 September 2025
- Urban Water Modeling Course (Graduate) at Florida Gulf Coast University: Remote Sensing as a Complementary Tool in Hydrologic Modeling - 18 November 2025

PhD students co-supervision (on going)

1. Weldamaryam, E.M. (2025). Assessing Climate and Land Use Changes Impacts on Water, Vegetation, and Soil, Mekelle University

MSc Students Supervision (completed)

1. Mihiret, A. W. (2022). Hydrogeological conceptualization and numerical modeling of groundwater resources in the Zamra catchment, Northern Ethiopia. https://essay.utwente.nl/86232/
2. Chekole, T. (2018): Hydrological flow of Ilala Catchment under Future climate and Different Land Cover Scenario
3. Desta, T. A (2017): Assessment of Groundwater Vulnerability to pollution Using SINTACS Model: - the Case of Aynalem Well field, Northern Ethiopia
4. Gebremichael, F. H (2017): Modelling urban expansion and Its Impact on nearby farmlands Using GIS and RS Techniques: the case of Wukro Town, Tigray and Northern Ethiopia.
5. Desta Y. B. (2016): Spatial and temporal drought analysis of Ragali watershed, western Danakil, northern Afar

Research Projects/Grants

1. Innovative Solutions for Improving Water Quality and Strengthening Local Economies in the Gulf of America Watershed ($999,590.00), EPA, No-cost collaborator (Awarded)
2. Impact of Conflict on Natural Resource and Sustainable Recovery Mechanisms (1,971,155.00 ETB), World Vision, Co-PI, 2024 (completed)
3. FSP (Euro 6,250.00), University of Twente, PI, 2023 (awarded)
4. Prediction of crop yield from space using machine learning models in Tigray, Ethiopia (493,680.00ETB), Mekelle University, CO-PI, 2020 (not awarded).
5. Hydrological response to land use and land cover change in Zamra catchment ($5,000.00), PoCFP, PI., 2019 (not awarded)
6. Hydrometeorological instrument establishment in Zamra catchment, Ethiopia ($15,000.00), International Foundation for Science, PI, 2019 (not awarded)
7. PhD study grant, €45,000.00, Dutch organisation for internationalisation in education (Nuffic) and University of Twente, 2018 (completed)
8. Estimation of spatial Groundwater Recharge Using GIS based WetSpass model, Northern Ethiopia (98,760.00 ETB), Mekelle University, PI, 2017 (completed)
9. Assessing Spatial distribution of aridity indices using GIS in Raya valley, northern Ethiopia (45,160.00 ETB), Mekelle University, PI, 2016 (completed)
10. The contribution of Environmental rehabilitation efforts to livelihood improvement of the rural community. The case of Medebay Zana wereda with Specific focus on honey production, irrigation and flood control (99,435.00 ETB), Mekelle University, Co-PI, 2017 (completed)
11. Geoheritage and Geodiversity development in the Danakil Depression, northern Ethiopia, Mekelle University, 185,600.00ETB, Co-PI (not awarded)

Conference Presentations

1. 24–26 Feb 2026: Mao, W., Yi, M., Cecil, M., Babushkin, P., Core, M., Elshall, A., Tziolas, N., & Gebremedhin, M. A. (2026). Assessing nutrient and pesticide dynamics from agricultural fields to lakes in South Florida through machine-learning models. UF Water Symposium, UF.
2. 24–26 Feb 2026: Babushkin, P., Mao, W., Cecil, M., Core, M., Yi, M., Elshall, A., Gebremedhin, M. A., & Tziolas, N. (2026). Numerical modeling of pesticide and fertilizer transport in South Florida. UF Water Symposium, UF.
3. 24–26 Feb 2026: Cecil, M., Yi, M., Babushkin, P., Mao, W., Core, M., Elshall, A., Gebremedhin, M. A., & Tziolas, N. (2026). *Impacts of chlorothalonil on ion concentrations in surface waters in the South Florida region*. UF Water Symposium, UF
4. 23 Jan 2026: Large language models for the democratization of groundwater and climate data: An example of groundwater climate resiliency in South Florida, 35th Annual Southwest Florida Water Resources Conference, FGCU
5. 19 Dec 2025: Quantifying Pesticide Transport Pathways Using a PWC Model and Global Sensitivity Analysis: A Case Study of Chlorothalonil in South Florida, AGU25 conference, oral presentation, https://agu.confex.com/agu/agu25/meetingapp.cgi/Paper/1932562
6. 17 Dec 2025: Impact of Future Rainfall and Sea-Level Rise on Groundwater-Surface Water Interaction and Salinity in South Florida, AGU25 conference, oral presentation, https://agu.confex.com/agu/agu25/meetingapp.cgi/Paper/1854064

7. 17 Dec 2025: Machine learning framework for red tide monitoring in Charlotte Harbor, West Florida Shelf, AGU25 conference, poster presentation, https://agu.confex.com/agu/agu25/meetingapp.cgi/Paper/1931731
8. 11 Dec 2024: The Impact of War on Natural Resources and Sustainable Recovery Mechanisms in northern Ethiopia, AGU24 conference, poster presentation https://agu.confex.com/agu/agu24/meetingapp.cgi/Paper/1671709
9. 13 Dec 2023: Armed Conflict in Tigray Sets Back Decades of Ecosystem Restoration Efforts, AGU2023 conference, https://agu.confex.com/agu/fm23/meetingapp.cgi/Paper/1420897
10. 04 Oct 2023: Spatio-temporal rainfall interception loss at the catchment scale from earth observation, ITC, University of Twente, Netherlands
11. 19–20 Dec 2019: Improving Satellite-Based Rainfall by Integrating with In-situ Data, Geoinformation Science and Earth Observation Annual Conference, Institute of Geoinformatics (I-GEOS), Mekelle University, Ethiopia
12. 24–25 Feb 2017: Agriculture and Environmental Management for Sustainable Development, College of Agriculture & Environmental Sciences, Bahir Dar University, Ethiopia
13. 28–30 Nov 2016: Impact of climate change on flood frequency analysis, International Conference on Sustainable Land and Watershed Management, Mekelle University & ECRC at EDRI, Mekelle, Ethiopia
14. 17 Nov 2016: Assessing spatial distribution of aridity indices, I-GEOS, Mekelle University
15. 10–11 May 2014: Quality of Education & Continuous Assessment, Adwa College of Teacher Education, Adwa, Ethiopia
16. 10–11 May 2014: ICT in Teaching & Learning, Adwa College of Teacher Education, Adwa, Ethiopia
17. 29–30 May 2010: Regional Conference, Adwa College of Teacher Education, Adwa, Ethiopia

Training and Workshops

1. 12-16 May 2025: Data Analysis with Artificial Intelligence Training, Dendritic Summer Academy on AI and Data Science, FGCU, USA
2. 18 Apr 2025: AI Integration in Teaching Workshop, FGCU, USA
3. 23 Mar 2025: Forum on Shaping the Future of Material Research and Education, Florida Gulf Coast University (FGCU)
4. 17–18 Mar 2025: Hurricane Ian Researchers Workshop, FGCU, USA
5. 01-05 Jan2025: Artificial Intelligence Fundamentals, Certificate of Nanodegree Program Completion, Udacity.
6. 4 Apr 2024: Dutch Earth Observation Day, Utrecht, Netherland
7. 7-8 Sep 2023: Assessing Ecosystem Health using Copernicus Marine Data, jointly organized by Copernicus, the Copernicus Marine Service, Mercator Ocean International and Gopacom.
8. 16 May 2023: PhD Ethics Workshop, ITC, Netherland
9. 13-15 Mar 2023: Hydrology studies in a Changing Climate, WEkEO
10. 04–17 Aug 2019: Joint Doctoral Summer School – Gender-Sensitive and Inclusive Academic Culture, VLIR-UOS Project, KU Leuven
11. 11 Jan 2019: Academic Integrity Workshop, University of Twente, Netherlands

12. 29–30 Nov 2018: TGS Introductory Workshop, University of Twente, Netherlands
13. 26–29 Nov 2018: Programming in R, Faculty of Geo-information Science and Earth Observation (ITC), University of Twente, Netherland
14. 17 May 20116: Introduction to Data Library (DL) in collaboration with Mekelle University and University of Colombia
15. 17–25 June 2016: ArcGIS Analysis, Geo-data Management, Geodatabase, Web & Mobile Mapping, Pan Africa Geoinformation Services PLC (ESRI Authorized Reseller), Addis Ababa, Ethiopia
16. 10–12 Jan 2015: Statistical Downscaling Model (SDSM) Training, University–Industry Community Linkage (UICL), Mekelle University, Ethiopia
17. 01–03 Jan 2015: SWAT Modeling Training, University–Industry Community Linkage (UICL), Mekelle University, Ethiopia
18. 06–17 Feb 2017: Geospatial Training Workshop – Secondary Cities Project, U.S. Department of State, Colorado State University, Mekelle University
19. 23 Jan 2018: Time Series Climate Data Analysis Using R Software, Institute of Climate and Society (ICS), Mekelle University & TU Dresden, Mekelle, Ethiopia
20. 14 Nov 2010: Certificate of Competence – Information Technology Assistance Level IV, Tigray TVET Bureau, Mekelle, Ethiopia
21. Dec 2008-June 2009: Higher Diploma in the Teaching Profession, Mekelle University, Ethiopia
22. 17 Mar 2007: Cisco Routing & Switching, PC Maintenance Training, ITSC Technology Support & Ethiopian ICT Development Agency, Addis Ababa, Ethiopia
23. 17 Mar 2007: Microsoft Server 2003 Management, Microsoft Certified Learning, ITSC Technology Support, Addis Ababa, Ethiopia

Technical Skills

- Integrated Hydrological Modeling
- Installation and Programming of hydrometeorological instruments
- Geophysical data acquisition and analysis using Electrical Resistivity Tomography (ERT)
- Remote Sensing
- Geographic Information System
- Geospatial Data Analysis
- Python and R Programing
- Machine Leaning
- Data Analysis
- Environmental Data Science
- Land Use Land Cover Dynamics
- Statistical Downscaling

Language

Fluent in English, Amharic and Tigringa

References

1. **M.W. Lubczynski (Maciek),** Professor of Hydrogeology, University of Twente**,** Faculty of Geo-Information Science and Earth Observation**,** Langezijds, 1134**,** Email: m.w.lubczynski@utwente.nl**,** Mob: +31534874277, The Netherlands
2. **Emiru Birhane Hizikias**, professor in Ecology, Mekelle University, P.o.box 390, Email: emiru.birhane@mu.edu.et / emibir@yahoo.com, Mob: +251914702336/ +4746398947, Mekelle, Ethiopia
3. **Ahmed S. Elshall**, Assistant Professor in Groundwater Hydrology, Department of Bioengineering, Civil Engineering, and Environmental Engineering, U.A. Whitaker College of Engineering, Holmes Hall, FGCU, 10501 FGCU Blvd. S., Fort Myers, aelshall@fgcu.edu, Mob: (239) 590-7591